

Aditya Bagri

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SUMMARY

AI researcher with end-to-end **medical imaging** experience; deployed wrist/hand X-ray **fracture detection & localization** models with clinical radiologists at 5C Network (**96.4% recall**, **0.60 AP**, 20K+ images); **ArXiv publication** on fracture detection; invited speaker at **Google DeepMind** on PaliGemma for medical imaging. Deep learning, computer vision, and statistical validation background across clinical and competitive ML settings. **AIR 63 / 6,500+** at Amazon ML Challenge. B.Tech CSAI, IIIT Delhi.

EDUCATION

Indraprastha Institute of Information Technology, Delhi
B.Tech, Computer Science and Artificial Intelligence

CGPA 7.81/10
Nov. 2022 – May 2026

TECHNICAL SKILLS

Core AI/ML: Deep Learning, Computer Vision, Medical Imaging AI, Machine Learning, Vision-Language Models (VLM), Natural Language Processing, Knowledge Distillation, Recommendation Systems
Frameworks: PyTorch, Hugging Face Transformers, Scikit-learn
Languages: Python, C++, Java, C, SQL
Agentic & LLM: LangChain, LangGraph, DSPy, MCP, RAG, ReAct, Tool-Calling, LLM-as-Judge, Self-Reflection, Evaluation / Observability / Tracing, Gemini, LoRA, QLoRA
Infra & Deployment: Docker, GCP, FastAPI, REST APIs, Git, Conda, Distributed / Federated Training

EXPERIENCE

AI Scientist Intern

May 2025 – Sept 2025

5C Network, Bengaluru

- Engineered a custom **SigLip**-based contrastive loss for fracture classification on Google's **PaliGemma** vision-language model, **improving recall by 19%** directly reducing clinically critical missed fractures in wrist/hand radiograph screening.
- Designed and **deployed** end-to-end wrist/hand X-ray **fracture detection & localization** models in collaboration with **clinical radiologists**, achieving **0.60 AP**, **96.4% recall**, and **91.07% accuracy** on **20K+** annotated images; results published on ArXiv.

Undergraduate Researcher

Aug 2025 – Present

MIDAS Lab, IIIT Delhi | Advisors: Dr. Avinash Anand, Dr. Rajiv Ratn Shah

- Designed **dKFD**, a phase-aware evidence-selection framework for long-video event understanding (anomaly detection in driving scenarios).
- Built an end-to-end **evaluation pipeline**: reproduced official DoTA VAR protocol with matched ablations and budget-scaling studies.
- Achieved **33.74% AVG Top-1**, surpassing the strongest published offline reference (**31.0%**); **submitted to ACCV'26**.

Research Consultant (Part-time)

Aug 2025 – Present

BITS-ON, IIIT Delhi | Advisor: Dr. Abhijit Mitra

- Built ML pipelines for ILA fault prediction in optical networks with a technical stakeholder, reaching **98% test accuracy** on real-world data.

PUBLICATIONS

- Bagri, A. et al. "Fracture Detection and Localization in Wrist and Hand Radiographs using Detection Transformer Variants." *arXiv*, 2025.

PROJECTS

Amazon ML Challenge 2025 – Multimodal Price Prediction | PyTorch, DeBERTa, ViT

Sept 2025

- Fine-tuned **DeBERTa-v3-Large** with custom Huber+L1+SMAPE loss and a gated regressor, ranking **AIR 63 / 6,500+**.
- Built a **multimodal text+image fusion** model with frozen DeBERTa, trainable **ViT**, and learnable gating, improving SMAPE from 44.112 to **43.451** — demonstrating end-to-end vision-language model fusion.

Distilled Preference Reasoning – LLM Recommender (Team of 5) | DistilRoBERTa, LLM, TMDB API

May 2026

- Built a **teacher–student recommender** on MovieLens-100K, **distilling** LLM-generated preference reasoning into a **DistilRoBERTa** ranking student — demonstrating knowledge distillation applied to a real ranking task; **led a team of 5**.
- Student reached **HR@10 0.94 / NDCG@10 0.78** on 6,223 queries vs 0.28/0.13 for MF-BPR baseline.

Multi-Agent Proof Verification Benchmark | Python, DSPy, Gemini, NVIDIA NIM

May 2026

- Built a **multi-agent** proof-verification framework in **DSPy** scoring **Gemini**-generated proofs across **7 LLM-as-judge** dimensions; added DSPy **prompt optimization** (GEPA/BootstrapFewShot) and a verifier-feedback **self-reflection / repair** loop.

ResearchOS – Agentic Research Synthesis System | LangGraph, LangChain, RAG, NVIDIA NIM

Feb 2026

- Built a modular **multi-agent** system in **LangGraph** with ReAct tool-use and hierarchical delegation, generating slide decks and notes across **10+ papers in under 5 min** with checkpointed pipeline, retries, and citation tracking.

ACHIEVEMENTS

- Invited speaker** at Google DeepMind's Open Models Meet-up to present research on **PaliGemma in medical imaging**.
- AIR 63 / 6,500+ teams** at Amazon ML Challenge 2025 (SMAPE 42.778) for Smart Product Price Prediction.
- Winner**, IdeaXhibit: Paper/Poster Presentation, Innerve 2026, for presenting own research work.